



Project Title: Wave-Based Computation Using Dynamic Liquid Media and Quantum Simulation

Team Name: Quantum Minds

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Team Members:

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Introduction to the Project

Problem Statement:

Develop a low-cost experimental setup to study wave propagation in liquids with dynamic obstacles. Capture the resulting wave patterns using a LiDAR sensor, extract the experimental data, and analyze it using Qiskit-based quantum computing techniques.

Objectives:

Design and build a low-cost experimental prototype. Generate liquid waves using sound. Create dynamic obstacles using ice cubes. Study the effect of different liquid viscosities on wave propagation. Capture and extract wave data using a low-cost LiDAR sensor. Analyze the extracted data using Qiskit.

Expected Outcome:

Develop a functional prototype for wave generation and sensing. Acquire real-time wave data using a low-cost LiDAR sensor. Extract meaningful features from the experimental data. Analyze the extracted data using Qiskit to explore quantum-based processing techniques. Evaluate how sound, viscosity, and dynamic obstacles influence wave behavior.

Working Principle

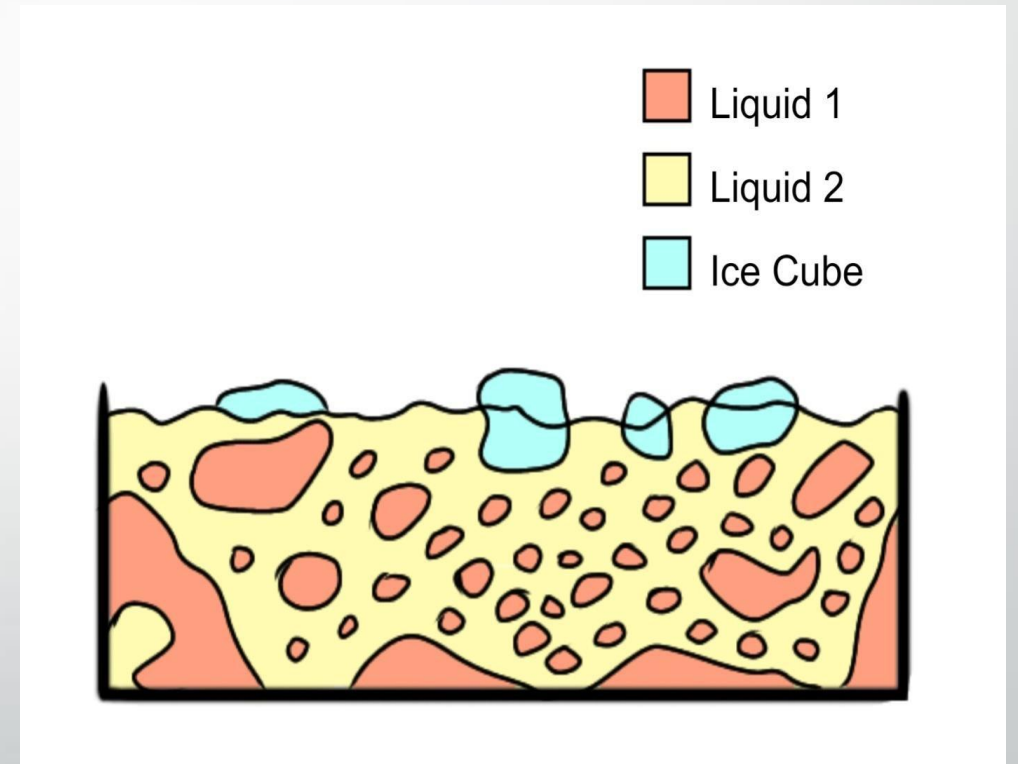
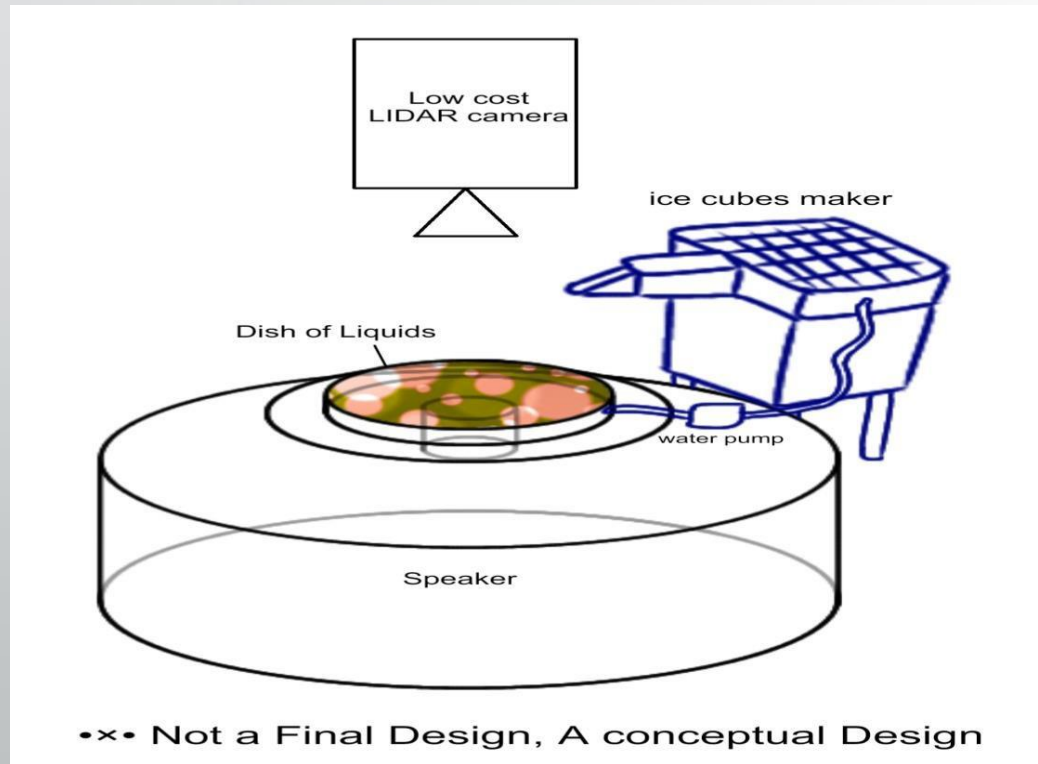
- Speaker generates controlled sound vibrations.
- Vibrations create surface waves in the liquid.
- Ice cubes act as temporary dynamic obstacles.
- Liquid viscosity influences wave speed and damping.
- Low-cost LiDAR continuously scans the liquid surface.
- Sensor data is processed to analyze wave patterns and obstacle interactions

Factors Affecting the System

Factors:

- **Waves:** Water in the dish is vibrated at high frequency to create complex wave patterns, which are detected using a low-cost LiDAR sensor.
- **Obstacles:** Ice cubes are introduced as temporary obstacles that gradually melt, creating a dynamic environment.
- **Sensor Noise:** Low-cost, high-sensitivity LiDAR sensors may introduce measurement noise, affecting prediction accuracy.
- **Viscosity:** Different liquids (water, glycerin mixtures, etc.) produce different wave propagation characteristics.
- **Sound:** Speakers generate sound waves that excite the liquid surface and create ripple patterns.
- **Temperature:** The liquids characteristics can change due to the temperature of the liquids and also affect the speed the ice(obstacle) melts.

Conceptual Design



Why Low-Cost LiDAR and Different Viscosity Liquids?

Low-Cost LiDAR

- Affordable
- Introduces Noise
- Easy integration
- Suitable for prototypes

Different Viscosity Liquids

- Change wave speed
- Produce different wave patterns
- Test system robustness

Why Ice Cubes?

Advantages:

- Dynamic obstacles
- Low cost
- Easy to replace
- Naturally change over time

Limitations:

- Requires additional cooling System
- Different liquids have different freezing points

Software Design

Physical System



LiDAR Data Collection



Signal Processing



Feature Extraction



Qiskit Simulation



Results & Analysis

Work Completed (Week 1)

- Finalized the project concept.
- Identified the key factors affecting the system.
- Studied sound-wave generation and ripple patterns.
- Selected a low-cost LiDAR sensor.
- Researched liquid viscosity effects.
- Analyzed ice cubes as dynamic obstacles.
- Started preparing the PPT and conceptual design.

Who Did What?

Team Member Contribution

- Harish Ganapathy S - Conceptual Design
- Ramya S - Literature Survey, LiDAR sensor research
- Akash S - Viscosity analysis and ice cube study

Next Week Goals

- Complete the conceptual design.
- Finalize the PPT.
- Develop the software architecture.
- Begin Qiskit simulation planning.
- Start implementation and testing.
- Collect additional references

Who Will Do What?

Team Member - Assigned Task

- Harish Ganapathy S - Finalize conceptual design and PPT, Prepare software architecture, documentation, and references.
- Ramya S - Complete LiDAR selection and hardware specifications.
- Akash S - Research suitable liquid mixtures and viscosity testing.

Conclusion

- The project concept has been finalized.
- Initial research and feasibility analysis have been completed.
- The conceptual design and system architecture have been prepared.
- The next phase will focus on hardware integration, software development, Qiskit simulation, and system implementation.
- We look forward to validating the proposed True Random Number Generator through experimentation and performance evaluation.



*Thank
you!*